

09_Error_Handling.md

Introduction

The Blogging System API shall implement centralized exception handling to provide consistent and secure error responses.

Error Handling Strategy

The application shall use Global Exception Middleware.

Responsibilities include:

- Catch unhandled exceptions.
 - Log exception details.
 - Return standardized JSON responses.
 - Prevent internal implementation details from being exposed.
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Standard Error Response

```
{
  "success": false,
  "message": "An unexpected error occurred.",
  "errors": []
}
```

Common HTTP Status Codes

Status	Meaning
200	OK
201	Created
204	No Content
400	Bad Request
401	Unauthorized

Status	Meaning
403	Forbidden
404	Not Found
409	Conflict
500	Internal Server Error

Validation Errors

Validation failures shall return HTTP 400 with descriptive messages.

Example:

```
{
  "success": false,
  "message": "Validation failed.",
  "errors": [
    "Email is required.",
    "Password must contain at least one special character."
  ]
}
```

Exception Categories

- Validation Exceptions
- Authentication Exceptions
- Authorization Exceptions
- Database Exceptions
- Unexpected Exceptions

Logging

Unhandled exceptions shall be logged through OpenTelemetry.

Sensitive information shall never be included in logs.

10_Project_Roadmap.md

Phase 1 — Documentation

- Project Overview
- Functional Requirements
- Non-Functional Requirements
- Architecture
- Database Design
- UML Diagrams
- API Specification

Status:  Complete

Phase 2 — Solution Setup

- Create Solution
 - Create Projects
 - Configure References
 - Install NuGet Packages
 - Configure Dependency Injection
 - Configure Swagger
 - Configure PostgreSQL
 - Configure Entity Framework Core
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Phase 3 — Authentication

- Registration
 - Email OTP Verification
 - Login
 - Refresh Tokens
 - Forgot Password
 - Reset Password
 - Change Password
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Phase 4 — User Module

- View Profile
 - Update Profile
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Phase 5 — Blog Module

- Create Post
 - Update Post
 - Delete Post
 - Drafts
 - Publish Draft
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Phase 6 — Comment Module

- Create Comment
 - Reply
 - Recursive Replies
 - Delete Comment
 - Lock/Unlock Comments
-

Phase 7 — Observability

- OpenTelemetry
 - Request Tracing
 - Exception Logging
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Phase 8 — Testing

- Build Verification
 - Database Testing
 - API Testing
 - Postman Collection
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Phase 9 — Finalization

- README Review
- Code Cleanup
- Documentation Review
- Final Git Tag